NextGen Institute

Machine Learning · Final Project Ideas

15 real-world problems · build a model, ship an app, and add it to your CV

The goal	Build a portfolio project you are proud to show employers: apply everything from the course — data understanding, preprocessing, EDA, modeling, tuning and evaluation — to a problem that a real company, hospital, school or farm actually has, and ship it as a live web app.
Why these ideas	Every idea uses **real company or public-sector data** and a problem that businesses pay to solve. All of them are solved with the classical ML from this course — regression, classification, clustering and PCA — no deep learning needed.
How to choose	Pick a field you care about and a problem your team finds exciting. Read the dataset card before deciding, and for very large datasets work on a well-chosen sample.
Rules	One idea per team unless the instructor approves. You may propose your own problem and dataset with a one-paragraph proposal for approval. Every number you report must come from your own code.

Ideas at a glance

#	Project	Field	ML task
1	**Late-Delivery Predictor for an E-commerce Marketplace**	E-commerce · Operations	Classification
2	**Customer Lifetime Value & Segmentation Engine**	Retail · Marketing	Clustering + Regression
3	**Will This Visitor Buy? Real-Time Purchase Intent**	E-commerce · Growth	Classification
4	**Store Sales Forecasting for a Retail Chain**	Retail · Planning	Regression
5	**Credit Risk Scoring for the Unbanked**	Finance · Lending	Classification
6	**Mobile-Wallet Fraud Detection**	FinTech · Payments	Classification (imbalanced)
7	**Supply-Chain Late-Delivery Risk Model**	Logistics · Supply Chain	Classification
8	**Ride ETA: Trip Duration Prediction**	Transport · Mobility	Regression
9	**Flight Delay Prediction**	Aviation · Operations	Classification
10	**Hotel Revenue Protection: Cancellation Predictor**	Tourism · Hospitality	Classification
11	**Diabetes Risk Screening Tool**	Healthcare · Public Health	Classification
12	**Hospital Readmission Risk**	Healthcare · Hospitals	Classification
13	**Jet Engine Remaining-Life Prediction**	Aerospace · Predictive Maintenance	Regression
14	**Semiconductor Production Yield: Fail Detection**	Manufacturing · Quality	Classification + PCA
15	**Electricity Demand Forecasting for the Power Grid**	Energy · Utilities	Regression (time series)

What every project must include

1 · Problem framing	Who has the problem, what decision the model supports, the target variable, and the metric that matters for the business (and why).
2 · Data understanding	Source, size, column dictionary, data types, missing values and class balance.
3 · EDA	Univariate, bivariate and multivariate analysis with clear charts, ending with 3–5 written insights or hypotheses.
4 · Preprocessing	Missing values, outliers, encoding, scaling and feature engineering inside a scikit-learn Pipeline. Split the data before fitting anything to avoid leakage.
5 · Modeling	A simple baseline (e.g., majority class or mean) plus at least three different algorithms from the course.
6 · Tuning & validation	Cross-validation and hyperparameter tuning (grid or random search) on the training data only.
7 · Evaluation	Final scores on the untouched test set with the right metrics (confusion matrix, precision/recall/F1, ROC-AUC, or MAE/RMSE/R^2), plus error analysis.
8 · Business insight	What the results mean in plain language and what the company should do next. Mention the model's limitations.
9 · Web app (required)	Save the final pipeline (joblib) and build a Streamlit or Gradio app that a non-technical user can use. Deploy it publicly on Streamlit Community Cloud or Hugging Face Spaces.

Deliverables

Notebook	One clean Jupyter notebook that runs top to bottom after Restart & Run All, with Markdown explanations between steps.
Presentation	10–12 slides, about 10 minutes: problem → data → key EDA insights → models → results → recommendation.
Live web app	**Required.** A deployed Streamlit or Gradio app with a public link, built around the app described in your idea card.
Repository	GitHub repo with a professional README: problem, live app link, screenshot or GIF, results table, how to run. Do not upload large datasets — link them.
LinkedIn post	A post presenting your project with a short demo video or screenshots and links to the app and repo (see the next page).

Suggested grading (100 points)

Criterion	Points
Problem framing & metric choice	10
EDA & insights	15
Preprocessing & pipeline (no leakage)	15
Modeling, tuning & cross-validation	15
Evaluation & error analysis	10
Business recommendation	10
Deployed web app (usable, public link)	15
Presentation, README & LinkedIn post	10

Make it portfolio-ready

A recruiter spends seconds on a project. These four things turn a course assignment into something worth showing.

1 · Live demo	A public app link anyone can open and try. A working app is the strongest proof that you can deliver, not just train a model.
2 · README	Start with one sentence on the problem, then the live link and a screenshot or GIF, then a results table (baseline vs. best model), then how to run it. Write it for a hiring manager, not for yourself.
3 · CV bullet	Formula: **action verb + what you built + data scale + measurable result + tools**. Every idea card has an example. Replace the [X] placeholders with your real numbers — never write a number you did not measure.
4 · LinkedIn post	Hook with the real problem → what you built → 1–2 key results → one thing you learned or a challenge you solved → 30–60 second demo video → links to app and GitHub → tag your teammates and NextGen Institute.

Example LinkedIn post structure

"Telecom companies lose customers every month and usually find out too late. In my NextGen Institute ML internship, our team built an app that predicts which customers are about to leave — before they do.

• Trained and tuned [N] models on 7,000 real customer records
• Our best model catches [X]% of customers who churn
• The biggest lesson: accuracy was misleading — recall is what the business needed

Try the live app: [link] · Code: [GitHub link]
#MachineLearning #DataScience"

Self-check before you publish

- ☐ App opens from a private browser window and handles wrong input without crashing
- ☐ README has the live link, a screenshot and a results table
- ☐ Notebook runs top to bottom with no errors
- ☐ Every number in the CV bullet and post comes from your own results
- ☐ The dataset is credited with its source link

The project ideas

01 Late-Delivery Predictor for an E-commerce Marketplace

E-commerce · Operations · Classification

The problem	A large online marketplace connects thousands of small sellers with customers. Late deliveries create bad reviews and lost customers. Predict, at purchase time, which orders will arrive late.
Why it matters	The company can warn customers, prioritize risky orders with carriers and spot sellers who are slowing down — the same problem Egyptian platforms like Noon and Jumia face.
Target	Late = delivered after the estimated delivery date (you create this label)
You will practice	Joining several related tables (orders, items, sellers, customers), building your own target, distance and time features, imbalanced classification.
The app (required)	Order form (product category, price, freight, seller and customer location) → late-delivery risk; a seller page showing each seller's late rate.
Stretch goal	Second model: predict a 1–2 star review from delivery and order features, and show how much late delivery hurts satisfaction.
Why it stands out	**Real company data spread over 9 tables — joining messy relational data is daily work for data scientists and rare in student portfolios.**
CV bullet	*Built a late-delivery prediction model on ~100K real marketplace orders across 9 joined tables, deployed as an order-risk app (recall [X]%).*
Dataset	**Brazilian E-Commerce by Olist (Kaggle, ~100K real orders, 9 tables)** https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce *Real anonymized commercial data released by the company — great for showing multi-table data skills.*

02 Customer Lifetime Value & Segmentation Engine

Retail · Marketing · Clustering + Regression

The problem	An online retailer treats every customer the same. Group customers by behavior and predict how much each one will spend in the next 3 months.
Why it matters	Marketing knows who to reward, who to win back and where to spend its budget — combining unsupervised and supervised ML in one product.
Target	Segments (unsupervised, RFM features) + next-90-day spend (regression, built with a time cut-off)
You will practice	Heavy cleaning (returns, cancellations, missing IDs), RFM feature engineering, K-Means vs. hierarchical, PCA for visualization, time-based target creation without leakage.
The app (required)	Customer dashboard: segments with their profiles; search a customer ID to see segment, predicted 90-day spend and the recommended action.
Stretch goal	Name each segment, attach a marketing action to it, and estimate the revenue at stake in each group.
Why it stands out	**Combines unsupervised and supervised ML in one product and speaks the language marketing managers use (segments, CLV, revenue at risk).**
CV bullet	*Built a customer segmentation and lifetime-value engine on ~1M retail transactions combining K-Means and regression, deployed as a marketing dashboard.*
Dataset	**Online Retail II (UCI, ~1M real transactions, 2009–2011)** https://archive.ics.uci.edu/dataset/502

03 Will This Visitor Buy? Real-Time Purchase Intent

E-commerce · Growth · Classification

The problem	Most website visitors leave without buying. Predict from a browsing session (pages visited, time spent, bounce rates, special days) whether the visitor will purchase.
Why it matters	Show a discount or a chat assistant only to visitors who are about to leave — more sales without giving discounts to everyone.
Target	Revenue (True / False)
You will practice	Session-level features, class imbalance, threshold tuning by business goal, comparing logistic regression, random forest and boosting.
The app (required)	Session form (pages, durations, bounce and exit rates, month, visitor type) → purchase probability and a suggested action (no action / show coupon / show chat).
Stretch goal	Simulate a campaign: if the site gives a coupon to the top 10% “likely to leave” visitors, how many extra sales could it get?
Why it stands out	**Ties the model directly to revenue — conversion and coupon ROI are questions every e-commerce company asks in interviews.**
CV bullet	*Developed a purchase-intent model for e-commerce sessions and deployed an app that recommends real-time actions to convert visitors (ROC-AUC [X]).*
Dataset	**Online Shoppers Purchasing Intention (UCI, ~12K sessions)** https://archive.ics.uci.edu/dataset/468

04 Store Sales Forecasting for a Retail Chain

Retail · Planning · Regression

The problem	A chain of over 1,000 stores must plan staff and stock six weeks ahead. Forecast daily sales per store using promotions, holidays, competition and store type.
Why it matters	Better forecasts mean fewer empty shelves, less waste and smarter staffing — a core problem for any supermarket chain.
Target	Sales (daily, per store)
You will practice	Merging store metadata, date and lag features, time-based validation (no random split), handling closed days, boosting models, RMSPE.
The app (required)	Pick a store and a date range → forecasted daily sales chart, with a promotion on/off toggle to compare scenarios.
Stretch goal	Explain which drivers move sales most (promotions, holidays, competitor distance) and turn them into planning advice.
Why it stands out	**A real Kaggle competition with a public leaderboard: you can state your score and compare it with thousands of professionals.**
CV bullet	*Forecast daily sales for 1,000+ retail stores with time-aware validation (RMSPE [X]) and deployed a what-if planning tool for promotions.*
Dataset	**Rossmann Store Sales (Kaggle competition, ~1M rows)** https://www.kaggle.com/c/rossmann-store-sales *Competition data: join the competition on Kaggle and accept its rules to download.*

05 Credit Risk Scoring for the Unbanked

Finance · Lending · Classification

The problem	Many people have little or no credit history, so banks reject them. Use application data and past loan behavior to predict who will repay — and give more people fair access to loans.
Why it matters	Financial inclusion with controlled risk: the same challenge Egyptian banks and microfinance companies face.
Target	TARGET (payment difficulties = 1)
You will practice	Large real data, many missing values, aggregating side tables into features, imbalance, ROC-AUC, model interpretability for a credit decision.
The app (required)	Loan application form → default probability, approve / review / decline decision and the main factors behind it.
Stretch goal	Turn predicted probabilities into an approve / review / decline policy and check fairness across age and gender groups.
Why it stands out	**One of the most famous credit-risk competitions; banks and fintechs instantly recognize it, and it proves you can handle large, messy data.**
CV bullet	*Built an explainable credit-risk model on ~307K real loan applications (ROC-AUC [X]) and deployed a loan-decision app with fairness checks.*
Dataset	**Home Credit Default Risk (Kaggle competition, ~307K loan applications)** https://www.kaggle.com/c/home-credit-default-risk *Start with the main application table, then add one side table at a time. Join the competition to download.*

06 Mobile-Wallet Fraud Detection

FinTech · Payments · Classification (imbalanced)

The problem	Mobile-money services (like Vodafone Cash or InstaPay in Egypt) are growing fast, and so is fraud. Detect fraudulent transfers and cash-outs among millions of transactions.
Why it matters	Stopping fraud protects customers and the provider, while false alarms block honest users — the model has to balance both.
Target	isFraud (1 / 0)
You will practice	Working with millions of rows (sampling strategies), extreme imbalance, engineering balance-change features, precision–recall trade-off, cost-based evaluation.
The app (required)	Transaction form or CSV upload → fraud score per transaction, plus a threshold slider showing missed fraud vs. false alarms and their cost.
Stretch goal	Build a cost matrix (missed fraud vs. blocked honest user) and choose the threshold that minimizes total loss.
Why it stands out	**Millions of rows and extreme imbalance — shows you can scale and choose metrics correctly, a key skill for fintech roles in Egypt.**
CV bullet	*Detected mobile-money fraud in millions of highly imbalanced transactions (PR-AUC [X]) and deployed a cost-based threshold tuning tool.*
Dataset	**PaySim Mobile Money Simulation (Kaggle, ~6.3M transactions)** https://www.kaggle.com/datasets/ealaxi/paysim1 *Simulated from real mobile-money logs, so patterns are realistic while customer data stays private.*

07 Supply-Chain Late-Delivery Risk Model

Logistics · Supply Chain · Classification

The problem	A global distributor ships orders across markets with different shipping modes. Predict which orders are at risk of late delivery before they ship.
Why it matters	Logistics teams can upgrade shipping mode or warn customers in time — directly relevant to logistics and export companies.
Target	Late_delivery_risk (1 / 0)
You will practice	Spotting and removing leakage columns (actual shipping days, delivery status), categorical encoding at scale, feature selection.
The app (required)	Order form (market, shipping mode, product, scheduled days) → late-delivery risk and a suggestion to upgrade shipping mode when risk is high.
Stretch goal	Add a second model that predicts order profit, and show which markets and shipping modes are both late and unprofitable.
Why it stands out	**Supply chain is a fast-growing hiring area; catching the leakage traps in this dataset is a strong talking point in interviews.**
CV bullet	*Built a supply-chain late-delivery risk model on ~180K global orders and deployed an app that recommends shipping upgrades for risky orders.*
Dataset	**DataCo Smart Supply Chain (Kaggle, ~180K orders)** https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

08 Ride ETA: Trip Duration Prediction

Transport · Mobility · Regression

The problem	Ride-hailing apps must show an accurate arrival time before the trip starts. Predict taxi trip duration from pickup and drop-off location and time.
Why it matters	Accurate ETAs build trust and drive pricing and driver dispatch — the core of apps like Uber, Careem and inDrive in Cairo.
Target	trip_duration (seconds, use a log transform)
You will practice	Geographic feature engineering (haversine distance, direction, location clusters with K-Means), time features, outliers, RMSLE.
The app (required)	Pick pickup and drop-off points on a map (or enter coordinates) and a time → predicted trip duration and the zone each point belongs to.
Stretch goal	Use K-Means on pickup points to discover city zones and show how travel time changes by zone and hour.
Why it stands out	**Geospatial feature engineering plus clustering shows creativity beyond fitting a model, and the problem is instantly clear to anyone.**
CV bullet	*Predicted ride durations from ~1.4M taxi trips with geospatial feature engineering and K-Means city zones (RMSLE [X]), deployed as an ETA app.*
Dataset	**NYC Taxi Trip Duration (Kaggle competition, ~1.4M trips)** https://www.kaggle.com/c/nyc-taxi-trip-duration *Join the competition to download; a 100K–200K row sample is enough for this course.*

09 Flight Delay Prediction

Aviation · Operations · Classification

The problem	Flight delays cost airlines and passengers time and money. Predict whether a flight will be delayed by more than 15 minutes using airline, airport, route and schedule.
Why it matters	Airlines can rebuild schedules, add buffer time on risky routes and inform passengers earlier.
Target	Delayed > 15 minutes (you create this label from ARRIVAL_DELAY)
You will practice	Big data (~5.8M flights — sample it), joining airline and airport tables, leakage (columns known only after departure), high-cardinality encoding.
The app (required)	Flight form (airline, origin, destination, month, day, departure hour) → delay probability and the safest departure time on that route.
Stretch goal	Rank airports and airlines by predicted delay risk and show which time slots are safest to fly.
Why it stands out	**Big public-sector data with a real leakage trap — explaining why you dropped columns shows senior-level thinking.**
CV bullet	*Built a flight-delay prediction model on U.S. flight records (ROC-AUC [X]) and deployed a delay-risk checker for travelers and planners.*
Dataset	**2015 Flight Delays and Cancellations (Kaggle, U.S. DOT)** https://www.kaggle.com/datasets/usdot/flight-delays

10 Hotel Revenue Protection: Cancellation Predictor

Tourism · Hospitality · Classification

The problem	Hotels lose revenue when guests cancel at the last minute. Predict whether a booking will be cancelled so the hotel can manage overbooking and deposits.
Why it matters	Smarter overbooking, deposit policies and revenue planning — directly relevant to Egypt's tourism sector.
Target	is_canceled (0 / 1)
You will practice	Detecting leakage (reservation status), date features, high-cardinality countries, comparing tree ensembles.
The app (required)	Booking form → cancellation probability and a recommended policy (no deposit / partial / full deposit).
Stretch goal	Recommend a deposit policy for each risk level and estimate the revenue it protects.
Why it stands out	**Clear money impact (protected revenue) in a sector central to Egypt's economy — easy to present to non-technical people.**
CV bullet	*Predicted hotel booking cancellations across ~119K real bookings and deployed a tool that recommends deposit policies by risk level.*
Dataset	**Hotel Booking Demand (Kaggle, ~119K real bookings)** https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand

11 Diabetes Risk Screening Tool

Healthcare · Public Health · Classification

The problem	Many people have diabetes or prediabetes and do not know it. Build a screening model that estimates risk from lifestyle and health survey answers.
Why it matters	A short questionnaire that tells people when to see a doctor — cheap, early detection at population scale.
Target	Diabetes_binary (0 = no, 1 = prediabetes or diabetes)
You will practice	Large real survey data, imbalance, choosing recall for screening, feature importance, calibration of predicted risk.
The app (required)	Short health questionnaire → diabetes risk level with the factors that raise it and a clear “not a diagnosis” note.
Stretch goal	Find the smallest set of questions (5–7) that keeps most of the model’s accuracy, to make a quick screening form.
Why it stands out	**Population-scale health data with a real-world constraint (a short questionnaire) — shows you design for users, not just accuracy.**
CV bullet	*Built a diabetes risk-screening tool from ~250K CDC survey responses, reducing the questionnaire to [N] key questions while keeping recall [X]%.*
Dataset	**CDC Diabetes Health Indicators (UCI, ~250K survey responses)** https://archive.ics.uci.edu/dataset/891 *A screening aid only — the app must clearly say it is not a medical diagnosis.*

12 Hospital Readmission Risk

Healthcare · Hospitals · Classification

The problem	Diabetic patients who return to hospital within 30 days are costly and a sign of poor follow-up. Predict which patients are at high risk at discharge time.
Why it matters	Hospitals can schedule follow-up calls and visits for high-risk patients, improving care and reducing costs.
Target	Readmitted within 30 days (Yes / No)
You will practice	Messy real clinical data (“?” as missing, diagnosis codes), grouping categories, imbalance, recall-focused evaluation.
The app (required)	Discharge form for a patient → 30-day readmission risk with the factors that raise it, for the care team to plan follow-up.
Stretch goal	Analyze which medications and admission types drive risk and present findings as recommendations for doctors.
Why it stands out	**Messy real clinical data and a recall-first decision — healthcare AI teams look for exactly this kind of judgment.**
CV bullet	*Built a 30-day readmission-risk model on ~100K real hospital encounters, cleaning messy clinical data, and deployed a discharge-risk tool.*
Dataset	**Diabetes 130-US Hospitals 1999–2008 (UCI, ~100K encounters)** https://archive.ics.uci.edu/dataset/296

13 Jet Engine Remaining-Life Prediction

Aerospace · Predictive Maintenance · Regression

The problem	Aircraft engines are maintained on fixed schedules, which wastes healthy parts or risks failures. Predict how many cycles an engine has left before it fails, from its sensor readings.
Why it matters	Maintenance at the right time: lower cost, higher safety — the same idea applies to factories, trains and power plants.
Target	RUL — Remaining Useful Life in cycles (you compute it from the data)
You will practice	Creating the target, rolling-window features from sensor time series, removing constant sensors, grouped train/test split by engine.
The app (required)	Pick an engine (or upload its sensor history) → predicted remaining cycles, a degradation chart and a maintenance alert.
Stretch goal	Turn the prediction into a maintenance alert (e.g., RUL < 30 cycles) and report precision and recall of the alert.
Why it stands out	**A NASA industry benchmark for predictive maintenance — a recognizable name on your CV and directly relevant to manufacturing and energy jobs.**
CV bullet	*Predicted the remaining useful life of jet engines from NASA sensor data (RMSE [X] cycles) and deployed a maintenance alert dashboard.*
Dataset	**NASA C-MAPSS Turbofan Engine Degradation (Kaggle)** https://www.kaggle.com/datasets/behrad3d/nasa-cmaps *Run-to-failure simulation data published by NASA — a classic industry benchmark.*

14 Semiconductor Production Yield: Fail Detection

Manufacturing · Quality · Classification + PCA

The problem	A chip factory records hundreds of sensor signals for every production run. Detect which runs will fail quality tests and find which signals matter.
Why it matters	Catching failures early saves expensive materials, and knowing the key signals tells engineers what to fix.
Target	Pass / Fail (-1 / 1)
You will practice	Very high dimensions (590 features, few rows), many missing values, feature selection vs. PCA, severe imbalance — a true test of the whole course.
The app (required)	Upload a production-run CSV → pass / fail prediction and a chart of the sensors that contributed most.
Stretch goal	Compare feature selection and PCA side by side, then report the 10 most important sensors to the engineering team.
Why it stands out	**590 features and very few failures — the hardest dataset here; succeeding with PCA and feature selection proves real depth.**
CV bullet	*Detected failing semiconductor production runs from 590 sensor signals using feature selection and PCA, and identified the 10 key sensors for engineers.*
Dataset	**SECOM (UCI, 1,567 production runs, 590 sensors)** https://archive.ics.uci.edu/dataset/179

15 Electricity Demand Forecasting for the Power Grid

Energy · Utilities · Regression (time series)

The problem	Power companies must know tomorrow's demand hour by hour. Under-forecasting causes blackouts and load shedding; over-forecasting wastes fuel. Forecast hourly electricity demand from past consumption and calendar patterns.
Why it matters	Accurate load forecasts keep the grid stable and cut fuel costs — the exact problem behind Egypt's 2023–2024 load-shedding hours.
Target	Hourly demand in megawatts (MW)
You will practice	Time features (hour, weekday, month, holidays), lag and rolling features, time-based split, comparing linear models with gradient boosting, MAPE.
The app (required)	Pick a date → 24-hour demand forecast chart with predicted peak hours highlighted and a capacity-limit warning.
Stretch goal	Forecast the next 24 hours and flag peak hours where demand is likely to exceed a chosen capacity limit.
Why it stands out	**Time-series forecasting on 10+ years of real grid data, linked to a problem every Egyptian has felt — a memorable story for any interview.**
CV bullet	*Forecast hourly electricity demand on 10+ years of real grid data (MAPE [X]%) and deployed a peak-load warning dashboard for grid planning.*
Dataset	**Hourly Energy Consumption — PJM U.S. grid (Kaggle, ~145K hourly records per region)** https://www.kaggle.com/datasets/robikscube/hourly-energy-consumption *Real grid operator data covering more than 10 years.*

Dataset links were checked in October 2026. Public datasets are sometimes moved or renamed; if a link stops working, search the dataset name on Kaggle or the UCI repository.